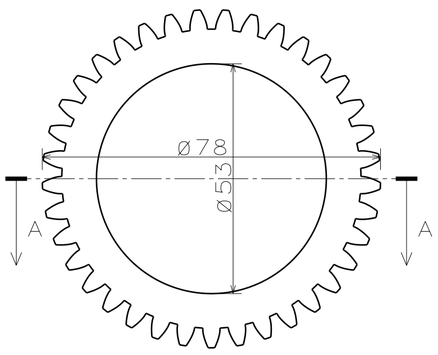
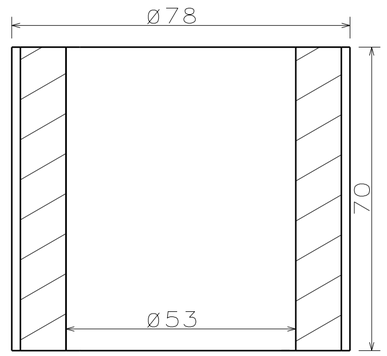


End view (1:1)



SectionA-A (1:1)

GEAR DATA / DISLI VERILERI (ISO 1328-1:2013; gear_accuracy_v1.csv, cad/params/planet.txt)
Number of teeth z: 37
Normal module m_n [mm]: 2
Helix angle [deg]: 0
Pressure angle [deg]: 20
Profile shift x: 0
Face width b [mm]: 70
Reference diameter d [mm]: 74
Tip / root diameter [mm]: 78 / 69
Bore = outer race of the planet bearing [mm]: 53
ISO 1328-1 class (profile class): 5 (4)
f_p / F_p [um]: 5.9 / 18
f_fa / f_Ha / F_a [um]: 4.3 / 3.4 / 5.5
f_fb / f_Hb / F_b [um]: 8.4 / 7.4 / 11
Runout F_r [um]: 16
Tooth thickness allowance E_sns / E_sni [um]: -40 / -65
Profile / lead modification tolerance [um]: 2
Flank finish: ground, superfinished
Mating gears: sun z45, ring z119
Dimensions on the views are the main sizes from the model; all other sizes are in the tables.

TOLERANCES AND CALLOUTS / TOLERANSLAR (tolerances_v1.csv; GD&T frames as this table, R19)
bore (outer raceway), datum A: cylindricity = 0.003 mm | -
teeth: radial runout (F_r) = 0.016 mm | A
matched set: tooth thickness: size spread within the set = 0.006 mm | -
teeth: ISO 1328-1:2013 class (pitch, helix / profile) = 5 / 4
teeth: tooth thickness allowance E_sns / E_sni (ring: space width) = -40 / -65 um

MATERIAL AND PROCESS / MALZEME VE PROSES (materials_v1.csv, P53)
Material: Pyrowear 53, VIM-VAR
Specification: AMS 6308
Condition: Carburized, hardened, sub-zero treated, double tempered at about 200-230 degC (above the 145-171 degC bulk temperatures of step 2.4)
Hardness and case: Surface 58-63 HRC; effective case depth to 550 HV after grinding 0.5-0.7 mm on the main gears (0.4-0.6 mm on the m 1.5 accessory gears; at most 0.4 m_n); core at least 30 HRC (STR-06); retained austenite at most 15% (TBR)
Surface texture: Flanks ground, then superfinished to Ra <= 0.15 um, Rz <= 0.8 um (STR-09); planet bore (outer race) ground and superfinished, Ra <= 0.1 um (TBR); roots protuberance-hobbed, not ground
NDT: Temper etch after grinding (ISO 14104 / AMS 2649), magnetic particle (ASTM E1444), case depth and core hardness on a test coupon of each heat-treat lot
Protection: Oil-wetted, no coating; preservation oil for storage

NOTES / NOTLAR
1. Case depth zone: teeth (flanks and roots) and the bore (integral outer race of the planet bearing) carburized as given under Hardness and case; faces and chamfers masked or ground after case hardening.
2. Surface texture as given under Surface texture; elsewhere Ra 1.6 um. 3. Unless noted ISO 2768-m; dimensions in mm; drawing per ASME Y14.5-2018 (metric) as the 3.8 basis.

PLANET GEAR P107 / PLANET DISLISI P107
Part / Parca: RGB-P107
Material / Malzeme: Pyrowear 53 (AMS 6308), carburized
Scale / Olcek 1:1 mm ISO 2768-m unless noted; ASME Y14.5-2018
Turboprop RGB, step 3.9 representative drawing / adim 3.9 temsili resim
Hamit Tanboga values: manufacturing/drawing_checklist_v1.csv
Sheet / Pafta 1/1 GD&T as the callout table (tolerances_v1.csv)